

SEDAR 13: Small Coastal Sharks, Finetooth Shark, Blacknose Shark, Atlantic Sharpnose Shark, and Bonnethead Shark

Data Workshop Research Recommendations

Life History Working Group

- Bonnethead life history in Atlantic Ocean, spanning the range of the stock.
- Re-evaluate finetooth life history in the Atlantic Ocean in order to validate fecundity and reproductive periodicity.
- Determine reproduction for finetooth in the Gulf of Mexico.
- Re-evaluate blacknose life history in Atlantic Ocean, spanning the range of the stock.
- Expand research efforts directed towards tagging of individuals in south Florida and Texas/Mexico border to get better data discerning potential stock mixing.
- Develop empirically based estimates of natural mortality.
- Coordinate a biological study for Atlantic sharpnose so that samples are made at least monthly, and within each month samples would be made consistently at distinct geographic locations. For example, sampling locations would be defined in the northern Gulf, west coast of Florida, the Florida Keys (where temperature is expected to be fairly constant over all seasons), and also several locations in the South Atlantic, including the east coast of Florida, South Carolina, and North Carolina. This same sampling design could be applied to all small coastal sharks.
- Population level genetic studies are needed that could lend support to arguments for stock discriminations using new loci and/or methodology that has increased levels of sensitivity.

Catch Histories Working Group

No research recommendations provided

Indices Working Group

The following recommendations provided in no particular order, deal with the collection of catch rate series data.

- Continuation of the fishery-independent surveys reviewed is encouraged. Some series that were not useful at this time may prove useful in the future with the inclusion of more data and series that were recommended for use at this time may improve with the additional information.
- If significant methodological changes are planned, it would be wise to have an overlap period between the gear, design, or vessel changes to all for calibration and quantification of those changes. This will allow for the time series to be maintained as one entity.

Assessment Workshop Research Recommendations

No research recommendations provided

Review Workshop Research Recommendations

The Review Panel chose to separate its report into several sections, starting with comments which pertained to all assessments, followed by discussion and recommendations for the small coastal shark complex and individual species assessed.

General research recommendations from the Data Workshop Report relevant to all species include the following

1. Re-evaluate life history in Atlantic Ocean, spanning the range of the stock.
2. Expand research efforts directed towards tagging of individuals in south Florida and Texas/Mexico border to get better data discerning potential stock mixing.
3. Develop empirically based estimates of natural mortality

Additionally, the following recommendations provided in no particular order, deal with the collection of catch rate series data.

The Review Panel encourages the continuation of the fishery-independent surveys reviewed. Some series that were not useful at this time may prove useful in the future with the inclusion of more data and series that were recommended for use at this time may improve with the additional information.

Small Coastal Shark Complex

With the development of species-specific data bases, SEDAR 13 used species-specific models for analysis. Nevertheless, for continuity purposes the species aggregated assessments were continued. However, it is the Review Panel's view that the aggregate analysis of the complex is unlikely to accurately reflect the status of every individual species in the complex and therefore it should not be viewed in isolation from the species-specific assessments. The aggregated results were not inconsistent with the assessment results on bonnethead and Atlantic sharpnose sharks, in particular. Therefore, the results of alternative forms of analysis were examined for differences and similarities in their structure and results, leading to advice on those species. This does not preclude that management of small coastal sharks as a complex may continue into the future; however, the scientific advice now focuses on the individual species within that complex. The Review Panel supports the Assessment Workshop decisions to provide assessment and advice on a species by species basis, rather than on the complex.

Finetooth Shark

Research recommendations from the Data Workshop Report are given above.

Additionally, the Review Panel has two more recommendations for finetooth shark. The first is to resolve the issue of negative r by targeted research on the life history of this species for both the Atlantic Ocean and the Gulf of Mexico. The second is to use an alternate model that is more appropriate to such a data-poor species. This class of model includes length- and stage-based density dependent matrix models or a delay-difference model. The assessment team is to be commended for endeavoring to apply more data-demanding models. However, the Review Panel is concerned that these models may give a misleading sense of confidence that isn't warranted.

Blacknose Shark

Research recommendations from the Data Workshop Report relevant to blacknose are given above.

Atlantic Sharpnose Shark

Recommendations are only made by the Data Workshop. Those of relevance to Atlantic sharpnose are as follows:

- a) Coordinate a biological study for Atlantic sharpnose so that samples are made *at least* monthly, and within each month samples would be made consistently at distinct geographic locations. For example, sampling locations would be defined in the northern Gulf, west coast of Florida, the Florida Keys (where temperature is expected to be fairly constant over all seasons), and also several locations in the South Atlantic, including the east coast of Florida, South Carolina, and North Carolina. This same sampling design could be applied to all small coastal sharks.
- b) Population level genetic studies are needed that could lend support to arguments for stock discriminations using new loci and/or methodology that has increased levels of sensitivity.
- c) Continuation of the fishery-independent surveys reviewed is encouraged. Some series that were not useful at this time may prove useful in the future with the inclusion of more data and series that were recommended for use at this time may improve with the additional information.

All three recommendations have merit but need to be judged on the basis of resources available and the priority/value of the fishery concerned. If the stock can be evaluated as not overfished and where no overfishing is occurring it is doubtful that increasing the level of sampling and research will change the effectiveness of management. It is also necessary to consider the opportunity costs of allocating resources to this species at the expense of other priorities. Recommendation (b) is only worthwhile if there is a capability to manage the two regions as separate stocks and that the fisheries operating in the two areas are sufficiently separate for this to make sense. For example, if vessels can transfer between areas, separate management may not be effective. A desk study using simulation models could be carried out to explore if a two stock approach is desirable, and if so, the more costly genetic study could be initiated.

With regard to (c), such surveys are often extremely costly and before an open ended commitment is made it would be desirable analyse the value of existing surveys and consider whether a more parsimonious approach might serve the purpose of the assessment without the need to support numerous surveys.

Bonnethead Shark

Research recommendations from the Data Workshop Report relevant to bonnethead sharks are given above in the general research recommendation section.

Comments Received Regarding the SEDAR Process

1. Evaluate the SEDAR Process. Identify any Terms of Reference which were inadequately addressed by the Data or Assessment Workshops; identify any additional information or assistance which will improve Review Workshops; suggest improvements or identify aspects requiring clarification.

The SEDAR process is a well thought out transparent consensus building process. Given the diversity of data and information sources, particularly for indices of stock size and biological parameters, putting the data together is a major task and it is appropriate to do so through a data workshop where all interested parties can participate. Similarly, analyzing the data through an Assessment Workshop whose tasks are to provide estimates of population parameters and trends as well as estimates of management benchmarks is appropriate. The Review Workshop, whose tasks are to evaluate the assessment methods and results and to provide the status declaration, with support from the assessment teams, provide an independent neutral evaluation of the methods, results and status determination.

The Data Workshop appears to have met the large majority of its terms of reference completely. Term of reference 3 was almost completely met, but the evaluation of how well the indices of stock size represented fishery and population conditions was not complete. For most stocks, at least some indices indicate conflicting trends over time, some increasing and some decreasing, while other indices were variable over time but showed no trends. The three conditions cannot adequately represent the conditions of the stock, assuming that the stock unit is appropriately defined, unless various geographical components of a stock complex behave differently over time. It is not clear if the selection of indices could be further refined at the Data Workshop or whether it would be more appropriately done at the Assessment Workshop, but it is clear that the selection of indices to be used in the modeling has to be further refined.

The Assessment Workshop appears to have successfully and completely met all its relevant terms of reference except that it did not provide research recommendations.

The process as implemented in SEDAR 13 could be improved by structuring the reports and the presentations more explicitly according to the terms of reference. It would also help to provide

more details of the exploratory runs, perhaps in a working paper so that the choice of final run can be better understood.

The review of finetooth shark assessment could have benefited by seeing the exploratory analyses of the life tables that were conducted by the assessment team who were very thorough. It would have given the Review Panel more confidence in the results from the input data.

Recommendations for future SEDAR assessments

Participants and the Review Panel commented throughout the week on the SEDAR assessment process. What follows is a non-prioritized list of the points made:

Sensitivity runs in the assessments should examine the robustness of stock status relative to the biological parameters that determine MSY. These include values for M, growth fecundity selectivity and the form of the stock recruitment curve.

Projection software tools should be developed that can incorporate uncertainty in the initial conditions and capture process error more comprehensively for the forecast period.

The Review workshop identified process error, especially in F, as a problem in determining stock status relative to MSY reference points. Further consideration needs to be given to a more robust means of interpreting stock status than the procedure of simply using the most recent data year. It is also important for managers to know the probability of exceeding reference points in the medium term, even if present stock status is judged satisfactory.

A more detailed and comprehensive analysis of the CPUE series would be desirable to evaluate the utility of many series available. A rigorous and objective scientific protocol should be developed against which CPUE series are evaluated as a basis for inclusion in assessments. This should include, *inter alia*, statistical design, spatial coverage and relevance to target species. The Review Panel envisioned a set of standards that delineated a weighted scoring depending on the attributes of the time series. For example, if the time series was based on a statistically valid sampling design targeted at the specific species, then it would achieve a high score for that standard. If the time series was properly designed for another species and largely covered the distribution in space and time, it would achieve an intermediate score against this standard, and so on. This would avoid vulnerability to personal preference and ad hoc choice of time series to include.

Differences between successive assessments, particularly when different data series or different assessment models are used, should be systematically investigated to assess whether differences are due to changes in data, changes in models, or changes in assumptions.